



DPUCANDBC

Reads messages from the CAN bus, or writes messages to it, whose format has been defined in a DBC file.

1. Inputs or Outputs:

(as defined in the DBC)

Values of the CAN message's fields which were last read (if Write=0);

Values of the CAN message's fields which will be sent next (if Write=1 or Write=2).

2. Static parameters:

CANMode

CAN bus channel over which the message is sent or re-ceived (see DPUCANBUS)

DBCFile

File name of the DBC file. If a relative path is entered, the file is searched below SILAB's installation directory (e.g. D:\SILAB).

ID

ID of the CAN message that is sent or received.

Extended

Defines if the CAN message is an extended message (true), i.e. whose ID is greater than 0xFFFF. The default value (false) is used for standard CAN messages.

MASK

Bit mask of the CAN message

Write

Defines if the message is written to the bus (1 or 2, see below) or read from the bus (0).

Period

If Write=1: Period (in ms), in which the data is written.

SendOnce

If Write=2: Whenever this signal rises from 0 to 1, the message is sent once.

Log

Record the data additionally to the CAN log file? (DPUCANLOG is required for the logging.)

3. Dependencies to other DPUs:

In addition to **DPUCANDBC**, a **DPUCANBUS** with the same *CANMode* must be running on the same computer, which is connected to the CAN hardware. The Index value of **DPUCANBUS** must be lower than the Index value of **DPUCANDBC**.

4. Functional principle:

This DPU provides inputs or outputs for a single CAN message which has been defined in a given DBC file. The inputs and outputs are named like the corresponding message channels in the DBC file.



5. Example for SendOnce:

The following example configuration can be used to send a message only once, whenever the driver passes over an event (hedgehog) on the road. (See the "Reference Database SCNX.pdf" for information about road events.)

```
DPUSCNXHedgehogKillerCANHK
{
    Computer={ASS};
    Index=10;
    Family="CAN";

    Name1="Send";
    OutMode1=(MonoFlop,500);
    Default1=0.0;
};

DPUCANDBCSendCAN
{
    Computer={CAN};
    Index=11;
    CANMode=33;
    DBCFile="lib\CANDBC\CANSend.dbc";
    Extended=1;
    ID=0x1ACCDDEE;
    Write=2;
};

Connections=
{
    CANHK.Out1->SendCAN.SendOnce
};
```

Whenever the hedgehog (...*CAN,Send,1*) is triggered, the output *Out1* of DPU *CANHK* is set to the value 1 used in the hedgehog for a period of 500 ms. Afterwards, it falls back to the default value (0.0), because *OutMode MonoFlop* was used in *DPUSCNXHedgehogKiller*. Thus, it is not necessary to place a second hedgehog in order to reset the value. The period of 500 ms makes it certain that the value can be transmitted to all computers that are part of the simulation.

This output is connected to the input *SendOnce* of the *DPUCANDBC*. Thus, the hedgehog sets the input *SendOnce* to 1 and then automatically back to 0. Afterwards, the message 0x1ACCDDEE is sent once using the current input values. These input values must be set before triggering the hedgehog by defining the other inputs of *DPUCANDBC*.